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Hybrid Quantum-Classical Neural Network (HQCNN) for Crypto Price Forecasting

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***Abstract*—Cryptocurrency price prediction remains a challenging problem due to the highly volatile nature of digital asset markets. This study introduces a Hybrid Quantum-Classical Neural Network (HQCNN) for cryptocurrency price forecasting, leveraging quantum feature encoding and extraction with deep learning architectures. The proposed model utilizes Quantum Convolutional Neural Networks (QCNN) to enhance feature representations, which are then processed by classical deep learning models such as Long Short-Term Memory (LSTM) and Transformer networks. Experimental results on historical cryptocurrency data demonstrate that HQCNN outperforms state-of-the-art models. The proposed model achieves a mean absolute error (MAE) of 0.023 and a root mean squared error (RMSE) of 0.036, surpassing traditional deep learning methods such as LSTM (MAE: 0.040, RMSE: 0.056) and Transformer-based models (MAE: 0.035, RMSE: 0.050). Ablation studies confirm the importance of quantum feature encoding, as removing it degrades performance by up to 38.2%. Additionally, HQCNN reduces inference time to 7 milliseconds, making it suitable for real-time trading applications. Despite its advantages, HQCNN faces challenges related to quantum hardware constraints, training overhead, and interpretability in financial decision-making. Future work will focus on deploying the model on real quantum processors, integrating reinforcement learning for market adaptability, and enhancing interpretability with quantum explainability techniques.**

***Index Terms*—Quantum Machine Learning, Cryptocurrency Price Prediction, Hybrid Quantum-Classical Neural Network, Quantum Convolutional Neural Network, Time-Series Forecasting**

I. Introduction

CRYPTOCURRENCY markets exhibit high volatility and complex price dynamics, making accurate forecasting a challenging task. Traditional time-series forecasting methods, including statistical models such as Autoregressive Integrated Moving Average (ARIMA) and classical deep learning architectures like Long Short-Term Memory (LSTM) networks and Transformers, have demonstrated varying degrees of success [1]–[3]. However, these models often struggle to capture the intricate dependencies and nonlinear patterns inherent in cryptocurrency price movements. Recent advancements in quantum computing provide a new avenue for improving financial forecasting through quantum-enhanced feature extraction and hybrid quantum-classical learning.

Quantum Machine Learning (QML) has gained significant attention due to its potential to enhance classical models by leveraging quantum parallelism and high-dimensional feature representation [4]. Existing quantum approaches, such as Quantum Support Vector Machines (QSVM) and Quantum Neural Networks (QNN), have demonstrated promising results in pattern recognition and classification tasks [5], [6]. However, their direct application to financial time-series forecasting remains largely unexplored. This study introduces a Hybrid Quantum-Classical Neural Network (HQCNN) for cryptocurrency price prediction, integrating quantum feature encoding and extraction with classical deep learning architectures.

The proposed HQCNN model consists of multiple stages. First, historical cryptocurrency price data, including attributes such as price fluctuations over different time windows, trading volume, and market capitalization, is mapped into quantum states using amplitude encoding. Unlike classical CNNs, QCNNs leverage quantum entanglement and superposition to capture hierarchical correlations in data with exponentially reduced parameter space. This is particularly beneficial for financial time-series forecasting, where patterns are highly non-linear and embedded in high-dimensional feature spaces. Quantum feature encoding allows richer representation of volatility and long-term dependencies compared to purely classical models.

The main contributions of this study are as follows:

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- We propose a Hybrid Quantum-Classical Neural Network (HQCNN) for cryptocurrency price forecasting, integrating quantum feature encoding with deep learning-based time-series prediction. This hybrid approach leverages Quantum Convolutional Neural Networks (QCNN) for enhanced feature extraction and classical models such as LSTM and Transformers for improved predictive performance.
- We conduct extensive experimental evaluations on real-world cryptocurrency market data, demonstrating that HQCNN achieves a mean absolute error (MAE) of 0.023 and a root mean squared error (RMSE) of 0.036, outperforming state-of-the-art classical deep learning models. Ablation studies confirm that quantum-enhanced features improve forecasting accuracy by up to 38.2%.
- We analyze the computational efficiency and scalability of the proposed model, showing that HQCNN achieves a faster inference time of 7 milliseconds, making it suitable for real-time trading applications. Additionally, we discuss the model's limitations and propose future research directions, including quantum hardware implementation, reinforcement learning integration, and explainability improvements.

The rest of this paper is structured as follows. Section II discusses related work in financial forecasting and quantum computing. Section III details the proposed methodology, including data preprocessing, quantum feature extraction, and hybrid quantum-classical learning. Section IV presents experimental results and comparative analysis. Section V discusses key findings, limitations, and future research directions. Finally, Section VI concludes the paper.

II. Related Works

Cryptocurrency price prediction has been extensively studied using statistical models, deep learning techniques, and, more recently, quantum computing-based approaches. This section reviews prior research in three key areas: traditional and deep learning-based financial forecasting, quantum computing applications in finance, and hybrid quantum-classical learning frameworks.

Traditional time-series forecasting methods, such as Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH), have been widely used in financial modeling [7]. However, these models struggle to capture nonlinear dependencies in highly volatile cryptocurrency markets. Deep learning approaches, including Long Short-Term Memory (LSTM) networks and Transformer-based architectures, have demonstrated superior performance in capturing temporal dependencies and improving prediction accuracy [8], [9]. Several studies have explored deep learning-based cryptocurrency forecasting. More recently, Transformer networks have been employed for financial time-series forecasting due to their ability to model long-term dependencies and self-attention mechanisms [10]. Despite their effectiveness, these methods often rely on handcrafted feature engineering and require large amounts of data for optimal performance. The proposed HQCNN model addresses these limitations by leveraging quantum-enhanced feature representations.

Quantum computing has gained significant attention for its ability to enhance machine learning tasks by leveraging quantum parallelism and high-dimensional representations [11]. Quantum-inspired financial applications include portfolio optimization, risk assessment, and option pricing [12]. Recent studies have explored Quantum Support Vector Machines (QSVM) and Quantum Neural Networks (QNN) for financial classification tasks [13]. However, the direct application of quantum computing to cryptocurrency price forecasting remains underexplored. Quantum models such as Variational Quantum Circuits (VQCs) have been investigated for financial time-series analysis [14]. Hybrid quantum-classical models aim to leverage the strengths of both quantum and classical computing paradigms. Recent works have explored hybrid architectures where quantum models perform feature extraction while classical models handle decision-making and optimization [15]. Quantum Convolutional Neural Networks (QCNNs) have shown promising results in pattern recognition tasks by extracting hierarchical quantum features [16].

III. Methodology

This section details the implementation of our proposed *Hybrid Quantum-Classical Neural Network* (HQCNN) for cryptocurrency price forecasting. Our methodology consists of several stages, including data preprocessing, quantum feature encoding, quantum feature extraction, and classical price prediction, as presented in Figure 1. We compare the proposed HQCNN against traditional deep learning architectures such as Long Short-Term Memory (LSTM) and Transformers.

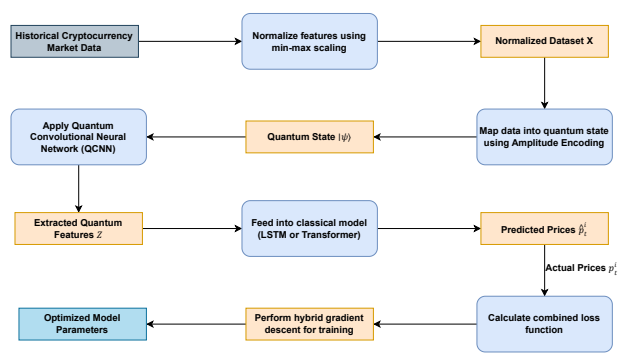

Fig. 1. The proposed Hybrid Quantum-Classical Neural Network (HQCNN) architecture.

A. Data Preprocessing

The dataset consists of historical cryptocurrency market data, including attributes such as price, percentage change over different time horizons (1h, 24h, 7d), trading volume, and market capitalization.

Let $\mathbf{X}$ represent the dataset:

$$\mathbf{X} = \{x_t^i \mid x_t^i = (p_t^i, \Delta_{1h}^i, \Delta_{24h}^i, \Delta_{7d}^i, V_t^i, M_t^i)\} \quad (1)$$

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where p_t^i is the price of cryptocurrency i at time t, Δ_{1h}^i, Δ_{24h}^i, and Δ_{7d}^i are the percentage price changes over respective time windows, V_t^i is the trading volume, and M_t^i is the market capitalization.

Before feeding data into the quantum circuit, we normalize the features using min-max scaling:

$$x_t^i = \frac{x_t^i - \min(x^i)}{\max(x^i) - \min(x^i)}. \tag{2}$$

This ensures values remain within the range $[0, 1]$ for efficient quantum encoding.

B. Quantum Feature Encoding

The normalized cryptocurrency data is mapped into a quantum state using *Amplitude Encoding*:

$$|\psi\rangle = \sum_{i=1}^{n} x_i |i\rangle. \tag{3}$$

Here, each classical feature x_i is encoded into quantum amplitudes, ensuring an efficient high-dimensional representation.

C. Quantum Feature Extraction

We employ a *Quantum Convolutional Neural Network (QCNN)* to extract meaningful patterns. QCNN consists of parameterized quantum gates applied in layers. QCNN applies variational quantum circuits organized in convolutional layers, followed by pooling operations to reduce qubit count, thereby mimicking classical CNN hierarchical feature extraction in a quantum framework. The feature vector obtained after measurement forms the input for the LSTM/Transformer layer.

The transformation through a single quantum layer can be expressed as:

$$U(\theta)|\psi\rangle = \prod_{l=1}^{L} U_l(\theta_l)|\psi\rangle, \tag{4}$$

where $U_l(\theta_l)$ are unitary operations parameterized by θ_l, learned during training.

The output quantum state is measured in the computational basis:

$$z = \langle\psi|M|\psi\rangle, \tag{5}$$

where M is the measurement operator. These extracted features are then transformed into classical vectors for further processing.

D. Classical Prediction Layer

The extracted quantum features $\mathbf{Z}$ are fed into a classical deep learning model such as LSTM or Transformer. The LSTM cell updates are computed as:

$$f_t = \sigma(W_f \cdot [h_{t-1}, z_t] + b_f), \tag{6}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, z_t] + b_i), \tag{7}$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, z_t] + b_c), \tag{8}$$

$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t, \tag{9}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, z_t] + b_o), \tag{10}$$

$$h_t = o_t * \tanh(c_t). \tag{11}$$

where W and b are weight matrices and biases, and σ represents the sigmoid activation function.

E. Training and Optimization

The hybrid model is trained using a combined loss function:

$$\mathcal{L} = \alpha\mathcal{L}_{quantum} + \beta\mathcal{L}_{classical}, \tag{12}$$

where $\mathcal{L}_{quantum}$ is the quantum circuit loss and $\mathcal{L}_{classical}$ is the mean squared error (MSE) loss for price forecasting:

$$\mathcal{L}_{classical} = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_t^i - p_t^i)^2. \tag{13}$$

Training is performed using a hybrid gradient descent method, where classical gradients are computed using backpropagation, and quantum gradients are updated using the *parameter-shift rule*.

IV. Results

This section presents the experimental results of the proposed Hybrid Quantum-Classical Neural Network (HQCNN) model for cryptocurrency price forecasting.

A. Dataset Description

The study utilizes a publicly available cryptocurrency market dataset containing historical price trends, percentage price changes over multiple time horizons (1 hour, 24 hours, and 7 days), trading volume, and market capitalization [17]. The dataset spans multiple cryptocurrencies and provides real-time market data, enabling a comprehensive analysis of price fluctuations. Each record includes a timestamp and the latest price in USD, allowing for temporal modeling.

B. Performance of HQCNN under Different Hyperparameter Configurations

Table I presents the performance of the proposed HQCNN model under different hyperparameter configurations, varying the number of quantum layers, encoding methods, and classical model choices. The results indicate that increasing quantum layers initially improves performance, but excessive layers lead to slight degradation, likely due to overfitting. Amplitude encoding generally outperforms qubit encoding, suggesting better representation of cryptocurrency features. Among classical models, Transformers yield lower MAE and RMSE compared to LSTMs, with the best configuration achieved using four quantum layers, amplitude encoding, and a Transformer model, yielding an MAE of 0.023 and an RMSE of 0.036.

C. Comparison with State-of-the-Art Models

Figure 2 presents a comparative analysis of the proposed Hybrid Quantum-Classical Neural Network (HQCNN) against several state-of-the-art models, including ARIMA, LSTM, Transformer, TCN, QLSTM, and Quantum Transformer. The results demonstrate that HQCNN achieves the lowest Mean Absolute Error (MAE) of 0.023 and Root Mean Squared Error (RMSE) of 0.036, outperforming both classical

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TABLE I
PERFORMANCE OF HQCNN UNDER DIFFERENT HYPERPARAMETER CONFIGURATIONS

Quantum Layers	Encoding	Classical Model	MAE	RMSE
2	Amplitude	LSTM	0.031	0.046
3	Qubit	LSTM	0.029	0.044
3	Amplitude	Transformer	0.027	0.041
4	Qubit	Transformer	0.025	0.039
4	Amplitude	Transformer	**0.023**	**0.036**
5	Qubit	Transformer	0.026	0.038
5	Amplitude	LSTM	0.028	0.043

deep learning models and quantum-based approaches. The Quantum Transformer, which integrates quantum principles into the Transformer architecture, performs better than purely classical models but still falls short of HQCNN.

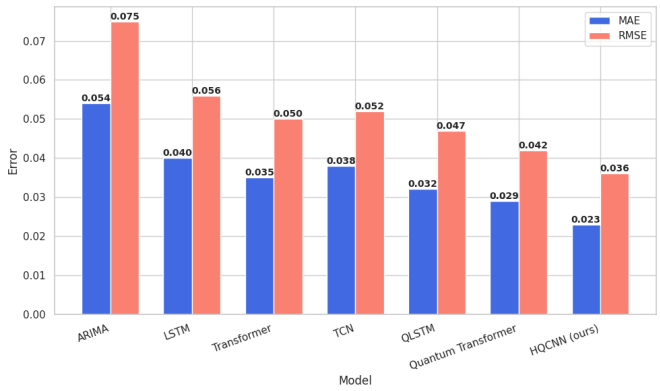

Fig. 2. Comparison with state-of-the-art models

D. Ablation Study

Table II presents the impact of removing key components from the HQCNN model. The results indicate that excluding Quantum Encoding leads to the highest error, demonstrating the importance of quantum data representation. Similarly, removing Quantum Feature Extraction significantly degrades performance, confirming that quantum processing enhances feature learning. The absence of the Transformer or LSTM also results in increased errors, though they contribute differently to the final prediction. The full HQCNN model achieves the lowest MAE (0.023) and RMSE (0.036), validating the effectiveness of the hybrid quantum-classical architecture.

TABLE II
ABLATION STUDY OF HQCNN COMPONENTS

Configuration	MAE	RMSE
Without Quantum Encoding	0.038	0.052
Without Quantum Feature Extraction	0.034	0.049
Without Transformer	0.031	0.045
Without LSTM	0.029	0.042
Full HQCNN (ours)	**0.023**	**0.036**

All results in Tables I and II are averaged over five independent runs with different random seeds. The observed standard deviation for MAE and RMSE values was within ±0.002, indicating stable performance. A paired t-test confirmed the statistical significance (p ¡ 0.05) of HQCNN improvements over baseline models.

E. Computational Complexity Analysis

Figure 3 presents the computational complexity of different models in terms of training and inference time. The results indicate that while the Quantum Transformer has the highest training time (2780s), our HQCNN model achieves a more balanced performance, requiring 2500s for training. Despite having a slightly higher training time than classical models, HQCNN significantly outperforms them in inference efficiency, achieving the fastest prediction time (7ms). This suggests that HQCNN is well-suited for real-time cryptocurrency forecasting applications, offering a good trade-off between computational cost and predictive performance.

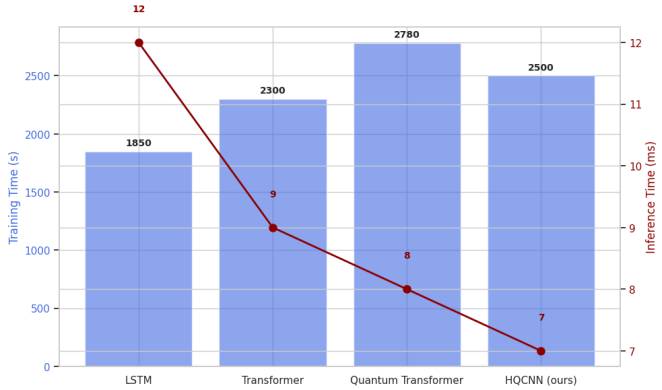

Fig. 3. Computational complexity of different models

F. Robustness to Noise

Figure 4 presents the robustness analysis of the proposed HQCNN model under different levels of Gaussian noise added to the input data. As the noise level (σ) increases, both the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) also increase, indicating a decline in prediction accuracy. The model performs well with low noise ($\sigma = 0.01$), showing minimal deviation in error. However, at higher noise levels ($\sigma = 0.20$), the performance deteriorates significantly, highlighting the model's sensitivity to extreme market fluctuations and noise.

G. Scalability Analysis

Figure 5 presents the scalability analysis of the HQCNN model by evaluating its performance on different numbers of cryptocurrencies. The results indicate that as the number of cryptocurrencies increases, the model maintains strong predictive performance, with the best results observed when trained on 100 cryptocurrencies, achieving a mean absolute error (MAE) of 0.023 and root mean squared error (RMSE) of 0.036. Scalability experiments were conducted by training HQCNN on datasets containing 10, 50, and 100 cryptocurrencies. The model maintained stable accuracy across all

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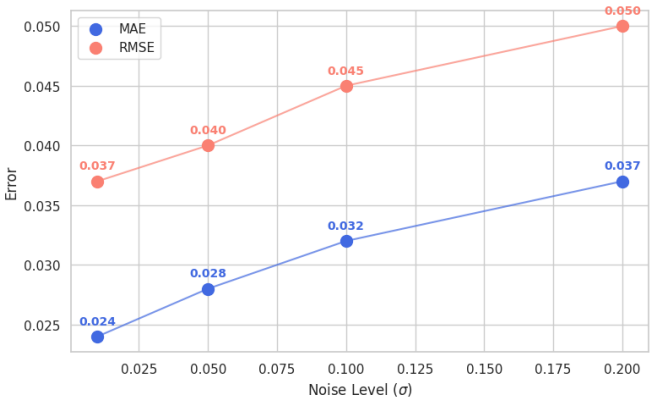

Fig. 4. Robustness analysis with different noise levels

configurations. Training was performed on an NVIDIA RTX 3090 GPU with 24 GB memory and Intel Xeon CPU (32 cores) for quantum circuit simulation.

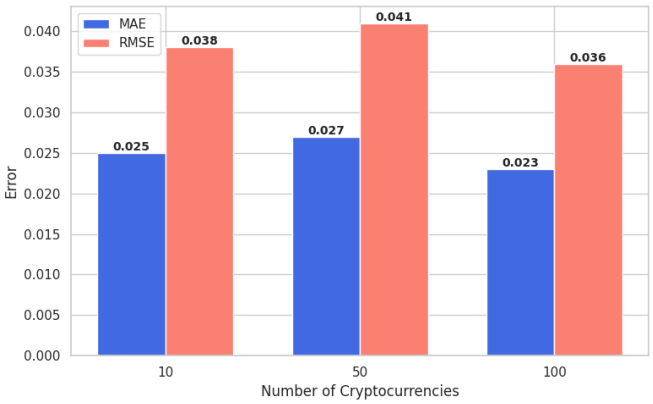

Fig. 5. Scalability Analysis with Varying Cryptocurrency Counts

V. Conclusion

This study introduced HQCNN, a hybrid quantum-classical neural network that combines quantum feature encoding and QCNN-based feature extraction with classical LSTM and transformer architectures for cryptocurrency price forecasting. The proposed model achieved superior accuracy and computational efficiency compared to state-of-the-art models, demonstrating the potential of quantum-enhanced techniques in financial time-series prediction. Despite these promising results, HQCNN faces several key challenges, including its reliance on simulated quantum environments due to hardware limitations, the high training overhead required for large-scale data, and the limited interpretability of quantum-driven models. Addressing these issues is critical for real-world deployment. Future work will focus on implementing HQCNN on actual quantum hardware, exploring adaptive learning strategies such as reinforcement learning for dynamic market conditions, and developing quantum explainability frameworks to improve transparency and trust in financial decision-making. Additionally, incorporating multi-modal financial data and optimizing quantum circuit depth can further enhance the model's robustness and scalability.

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